INCH-POUND
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SUPERSEDING
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15 November 1992

COMMERCIAL ITEM DESCRIPTION

LACES, FOOTWEAR

The General Services Administration has authorized the use of this Commercial Item Description for all Federal agencies.

1. SCOPE. This Commercial Item Description covers technical requirements for various nylon laces intended for use in footwear and equipage items utilized by Department of Defense, and other Government agencies.

2. CLASSIFICATION. The laces will be provided in the following types and classes:

2.1 Types and Classes.

Type I: Spun nylon laces

Class 1 - 7/32-inch width, flat
Class 2 - 5/16-inch width, flat
Class 3 - 9/32-inch width, flat

Type II: Bulked filament nylon laces

Class 1 - 3/32-inch diameter, round (210 denier yarn)
Class 2 - 1/4-inch width, flat
Class 3 - 1/8-inch diameter, round
Class 4 - 3/32-inch diameter, round (420 denier yarn)

Type III: Multifilament nylon laces

Class 1 - 1/8-inch each side, square

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AMSC-N/A FSC 8335

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3. SALIENT CHARACTERISTICS. The laces shall be made of 100 percent braided nylon with attached tips. The laces shall conform to the characteristics and requirements in Table I and Table II and meet the requirements of this document when tested as outlined in Table III.

3.1 Physical characteristics. Type I, II and III laces, with corresponding classes, shall conform to the physical characteristics specified in Table I.

TABLE I. Physical characteristics

Type/Class	Yarn type	Braid	Stuffer core
Type I Class 1, 2 and 3	Spun staple twisted into 2-ply yarn	Tubular without a core; two (2) over and two (2) under braiding, pressed flat	-
Type II Class 1	210 denier continuous multifilament increased to 230-250 denier by air bulking process	Tubular without a core; two (2) over and two (2) under braiding, round, shall not be pressed	-
Type II Class 2	210 denier continuous multifilament increased to 230-250 denier by air bulking process	Tubular without a core; one (1) pair over and one (1) pair under or one (1) end over and one (1) end under braiding, pressed flat	-
Type II Class 3	Cover and warp yarns 840 denier high tenacity continuous multifilament yarn increased to 940-1000 denier by air bulking process	Tubular with stuffer core; two (2) over and two (2) under braiding, round, shall not be pressed. Stuffer core not braided	10,080 denier high tenacity continuous multifilament nylon yarn(not air bulked or braided)
Type II Class 4	420 denier continuous multifilament yarn increased in size to 470-490 denier by air bulking process	Tubular without a core; two (2) over and two (2) under braiding, round, shall not be pressed	-
Type III Class 1	1050 denier by three (3) cabled yarns manufactured from high tenacity, continuous multifilament yarns, 1/	Square braid exhibiting four (4) separate sides with equal number of picks per inch and equal width per side of 1/8 (±1/32) inch, shall not be pressed	-

1/ The individual yarns used for Type III, Class 1 nylon laces, shall be twisted initially with not less than seven (7) nor more than nine (9) turns per inch (TPI) and the final plied twist shall be in the opposite direction with not less than 13 and not more than 15 turns per inch.

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3.2 Physical requirements. The braid for the laces (all types and classes) shall conform to the physical requirements specified in Table II, when tested as specified in Table III.

TABLE II. Physical requirements.

Type	Class	Width Tolerance (± 1/32-inch)	Carriers (min.)	Ends per carrier (min.)	Warp ends (min.)	Picks per inch (min.)	Breaking strength, lbs. (min.)	Stuffer (min.)
I	1	7/32 inch	44	1	-	38	95	-
I	2	5/16 inch	44	2	-	26	175	-
I	3	9/32 inch	44	2	-	21	185	-
II	1	3/32 inch (diam.)	16	1	-	28	100	-
II	2	1/4 inch	44, 1/	1 or 2, 2/	-	17	190	-
II	3	1/8 inch (diam.)	8	2	4	16	180	1
II	4	3/32 inch (diam.)	16	2	-	24	135	-
III	1	1/8 inch (each side)	8	1	-	14	250	-

1/ Carriers braiding in pairs or braiding as a basket weave.
2/ Only one (1) end per carrier shall be required when braiding one (1) end over and one (1) end under.

3.3 Lace tipping. Both ends of each nylon lace shall be provided with tips 3/4 (± 1/16) inch long. Types I and II tips shall be made from nylon or acetate sheet or film having a thickness of 0.013-0.016 inch when tested in accordance with ASTM D1777. As an alternative, the ends of Types I and II nylon laces shall be self-tipped by an ultrasonic or heat molding process. The ends of Type III, Class 1 nylon laces shall be self-tipped using an ultrasonic or heat molding process. Lace tips shall have sufficient rigidity to perform their intended function. The color of the tips shall be black or clear for black laces, and clear for all other laces.

3.4 Length. The length of the laces including tips shall be as specified in the solicitation. For laces less than 40-inches the tolerance shall be (+5.0/-0) percent. For laces 40-inches and longer the tolerance shall be (+3.0/-0) percent when tested in accordance with 3.4.1.

3.4.1 Length test. The finished laces shall not be stretched or placed under tension, prior to the length test. The finished lace shall be suspended vertically under a load of 3-ounces for Classes 1 and 4. The finished lace shall be suspended vertically under a load of 6-ounces for Classes 2 and 3. Then measure the lace (all classes) from end to end, including tips.

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3.5 Width. The width of the laces shall be determined at three (3) points at least 9-inches apart on each specimen in the sample unit. The measurements shall be taken by steel ruler graduated in thirty-seconds of an inch or by a caliper-type gauge. Measurements shall be taken with the specimen suspended vertically under a load of 3-ounces for Classes 1 and 4. Measurements shall be taken with the specimen suspended vertically under a load of 6-ounces for Classes 2 and 3. The results of all determinations on the sample unit shall be averaged and reported to the nearest 1/32-inch. The results shall be recorded as pass or fail.

3.6. Material and end item tests.

TABLE III. Requirements and test methods.

Characteristics	Requirement	Test method
Fiber Identification	3.	ASTM D276 or AATCC 20
Yarn Type	Table I	Visual
Turns Per Inch (TPI)	Table I (footnote)	ASTM D1423
Width	Table II	3.5
Carriers	Table II	Visual 1/
Ends per carrier	Table II	Visual 1/
Warp ends	Table II	Visual 1/
Picks per inch	Table II	Visual 1/
Stuffer	Table II	Visual 1/
Breaking strength	Table II	3.7
Thickness (lace tipping)	3.3	ASTM D1777
Length	3.4	3.4.1
Colorfastness to water (min)	3-4 (rating)	AATCC 107 2/, 3/
Color	3.8	3.8.1
Toxicity	3.9	3.9.1

1/ One determination per sample unit.
2/ The time shall be 1-hour
3/ Shall be rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change; AATCC Evaluation Procedure 2, Gray Scale for Staining.

3.7 Breaking Strength. To determine the breaking strength the following test method shall be followed:

a. Sample Preparation:
   1) Test specimens shall be a single length of the finished cordage not less than 24-inches long.
   2) Unless otherwise specified in the procurement document, five (5) specimens shall be tested from each sample unit.

b. Apparatus:
   The apparatus for determining the breaking strength shall be as described in ASTM Test Method D5034, except that the flat clamps shall be replaced with smooth clamps of the spool or drum type and the speed of the machine before the application of a

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load to the specimen shall be 6 (± 1)-inches per minute.

c. Procedure.

Prepare sample as specified above. Unless otherwise specified in the procurement document material testing shall be as specified below: The distance between the centers of the clamps (drum-type) at the start of the test shall be 6 (± 1/16)-inches. When the lace length is less than 24-inches a flat-type jaw shall be used in lieu of a drum-type jaw. The specimen shall be placed in the clamps or drums of the machine and a load applied until the specimen ruptures.

NOTE: If a specimen slips between the jaws, breaks in the jaws for any reason of machine malfunction an individual measurement falls markedly below the apparent average test results, such a measurement shall be disregarded and another specimen shall be tested from the same sample unit.

d. Calculation:

The breaking strength shall be the average of the specimens tested and shall be reported to an accuracy of 0.5 percent of the indicated (recorded) breaking strength. Each individual value used to calculate the average for breaking strength shall also be reported.

3.8 Color. The color of the laces shall be as specified in the contract or solicitation (see 7.3). Stuffer yarns for Type II, Class 3 laces may be undyed.

3.8.1 Visual shade matching. The color and appearance of the laces, shall match the standard sample (see 7.4) when viewed using AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200) K, with illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

3.9 Toxicity. The finished laces shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 3.9.1. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.9.1 Toxicity test. When required (see 7.3), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the laces are not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.4). If the toxicity requirement (see 3.9) can be demonstrated with historical use data, toxicity testing may not be required (see 7.3).

3.10 Labeling. Each set of laces shall have a label, adhered to the item and clearly legible. The label shall be printed and include information related to length of laces, color, and name of manufacture.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 End item examination. The visual and dimensional examination for defects shall be in accordance with ANSI-ASQ Z1.4 (see 7.2.2).

5.3 Visual examination. Each lace shall be examined for the following defects as specified in Table IV. The sample unit is one (1) lace.

TABLE IV. Visual examination defects.

Material defects Laces: Yarn - any broken, frayed yarns, or yarns exhibiting rubs or abrasion marks; any hole, cut, or tear. Any pulled yarns or uneven tension resulting in uneven width. Braiding - any uneven braiding resulting in open places or breaks in continuity; other than tubular braided with core, when applicable; flat lace, when applicable, not flat for a distance of 2-inches or more; round lace, when applicable, not round for a distance of 2-inches or more; number of yarns or type of braiding not in compliance with Table I; any slub, knot, or abruptly thickened place affecting utility. Color other than specified, offshade. Cleanness - any spot, stain, or other discoloration affecting appearance; objectionable odor. Any dimension not as specified or outside the tolerance. Lace tipping: Any tip missing, damaged, malformed, or insecurely attached. Inadequate heat molding (when applicable), e.g. burned, brittle, incomplete, sharp edges. Tipping of braid not neatly accomplished. Color of tips not as specified. End of tip beyond the braid more than 1/16-inch (for attached tips only); end of braid projects beyond the tip end more than 1/16-inch (for attached tips only). Thickness, width, or diameter of tip greater than thickness,
Labels: Label(s) missing, incorrect or missing information, or illegible.
Packaging: Not packaged in accordance with the contract or purchase order

5.3.1 Dimensional examination. The finished laces shall conform to the dimensions specified in the solicitation. Any dimension that is not within the established tolerance (see 3.4) shall be classified as a defect. The sample unit shall be one (1) lace.

5.4 Inspection requirements. Bid samples may be required for nylon footwear laces to assure product quality.

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5.5 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.3).

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking are as specified in the contract or order (see 7.3).

7. NOTES.

7.1 Sources of Government documents.

7.1.1 Copies of Government documents are available online at http://quicksearch.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.1.2 Federal Acquisition Regulations are available online at http://acquisition.gov/far/index.html or by contacting the Superintendent of Documents at 202-512-1800.

7.2 Sources of Non-Government documents.

7.2.1 AATCC Test Methods are available online at http://www.aatcc.org or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.

7.2.2 American Society for Quality (ASQ) documents are available online at http://www.asq.org or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.

7.2.3 ASTM Standards are available online at http://www.astm.org or from ASTM INTERNATIONAL, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.

7.2.4 Principles and Methods of Toxicology, A Wallace Hayes (editor), is available online from http://www.taylorandfrancis.com or from Taylor and Francis, 7625 Empire Drive, Florence, KY 41042-291.)

7.2.5 The contracting activity may elect to accept a certificate of compliance for the lace tipping thickness (see 3.3).

7.3 Ordering data. The contract or order should specify the following.

a. Commercial Item Description number and revision.
b. Type and Class of lace required (see 2.1).
c. Color required (see 3.8).
d. Length required (see 3.4).
e. When toxicity testing is required (see 3.9, 3.9.1).
f. Acceptance criteria (see 5.5).
g. Packaging requirements (see 6.1).

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7.4 Standard samples

7.4.1 For standard samples (if applicable), address the contracting activity issuing the invitation for bids or request for proposal.

7.5 Subject term (key word) listing.

Boots Braided
Equipage
Multifilamen
Nylon
Shoes Yarn

MILITARY INTERESTS:

Custodians:
Army – GL
Navy – NU
Air Force – 11

Review Activities:
Army – MD
Navy – CG
Navy – MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:
DLA-CT
Agent: Army-GL

Project Number: 8335-2016-001

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